No-arbitrage for lift lending

First-principles cycles from pallet math

Subtensor lending (PR #3268)
Working note — rewrite (pallet-first; Sept 2026 cross-check)

2026-10-29

Abstract

We derive the TAO-pure lending cycle **open short → pump → close → unstake** from the implemented pallet math (lift fraction, Balancer weights, invariant, force-sell, readd). Fees are an explicit additive parameter, set to zero in the base case. The lift fraction is denoted φ and is **not** a fee. The buggy close path (readd via weight re-derivation at the post-pump mark) predicts the clone SN29 drains (+289 TAO / +122 TAO). Weight-preserving readd flips the sign. Under keep-weights we state an explicit bounded-search pass/fail criterion (no $\leq N$-action self-contained sequence yields attacker net TAO > 0 for φ up to the side-cap) and report results on live reserves. Lifted liquidity deliberately shallows the book: slippage scales as $\approx 1/(1-\varphi)$ while a loan is open. μ's September 2026 note is cited only as a cross-check after our closed forms are in hand.

Contents

1 Notation **2**

2 Balancer and lift primitives **2**
2.1 Swaps (stake / unstake) . . . 2
2.2 Lift . . . 3
2.3 Force-sell (short open) . . . 3
2.4 EMA band and LTV (feasibility only) . . . 3
2.5 Readd . . . 3

3 Cycle: short → pump → wallet repay → unstake **3**
3.1 Base case $\varepsilon = 0$ . . . 3
3.2 Attacker PnL . . . 4
3.3 Production fee . . . 4

4 Prediction of the SN29 clone **4**
4.1 Two-wallet (a13) . . . 4
4.2 Single-wallet (a15) . . . 4

5 Root fix **5**

6 Pass/fail criterion (keep-weights) **5**

7 Deliberate volatility: slippage $\approx 1/(1-\varphi)$ **6**

8 Cross-check: μ, September 2026 **6**

9 Fixed weights are not Uniswap v2 **7**

1 Notation

All symbols below are defined from the live pallet / swap code (`pallet-lending`, `pallet-subtensor-swap`), not from an external handout.

Symbol	Unit	Meaning
x	α	`SubnetAlphaIn` (base reserve)
y	τ	`SubnetTAO` (quote reserve)
w_b	—	Balancer **base** (alpha) weight
w_q	—	Balancer **quote** (TAO) weight; $w_b + w_q = 1$
L	—	Invariant $L = x^{w_b} y^{w_q}$
p	τ/α	Spot $p = (w_b/w_q)\,(y/x)$
φ	—	**Lift fraction**. Short: $\varphi = P/x$ via `lift_fraction` (returns (P, x)). Long: $\varphi = P/y$.
P	α or τ	Principal (short: alpha; long: TAO)
λ	—	LTV (short default 0.25)
C	τ or α	Collateral
β	—	EMA band (default 0.15)
ε	—	Swap fee on **input** (production 33/65535; **base case** $\varepsilon = 0$)
S	τ	Pump size (`AddStake` / buy alpha)
B	τ	Post-pump TAO spent to buy repay alpha
E	τ	Short escrow proceeds from force-sell
F_{cyc}	—	Attacker terminal/initial TAO (profit iff $F_{\text{cyc}} > 1$)

Remark 1 (Clone report "w_1"). The SN29 clone report prints a field `w1` with $L \approx x^{1-w_1} y^{w_1}$. That printed w_1 is our w_q (quote / TAO). We never use w_1 in this note.

2 Balancer and lift primitives

2.1 Swaps (stake / unstake)

At fixed (w_b, w_q), fee ε on input:

$$\text{sell } \Delta x: \quad \Delta y = y\left(1 - \left(\frac{x}{x + \Delta x(1-\varepsilon)}\right)^{w_b/w_q}\right), \tag{1}$$

$$\text{buy with } \Delta y: \quad \Delta x = x\left(1 - \left(\frac{y}{y + \Delta y(1-\varepsilon)}\right)^{w_q/w_b}\right). \tag{2}$$

Weights are **not** updated mid-swap.

2.2 Lift

`lift_protocol_liquidity`(φ) removes $(\varphi x, \varphi y)$ with weights unchanged (`pallets/swap/.../lift_protocol_liq`

$$(x, y, L) \mapsto ((1-\varphi)x, (1-\varphi)y, (1-\varphi)L), \qquad p \text{ unchanged.}$$

For a short, pallet sets $\varphi = P/x$ so the alpha leg equals principal.

2.3 Force-sell (short open)

Sell $P = \varphi x$ into the post-lift book. At $\varepsilon = 0$, x returns to its pre-lift value and

$$y_s = y\,(1-\varphi)^{1/w_q}, \qquad E = (1-\varphi)y - y_s = y\Big[(1-\varphi) - (1-\varphi)^{1/w_q}\Big]. \tag{3}$$

(Proof: post-lift reserves $((1-\varphi)x, (1-\varphi)y)$; sell φx gives $y_s = (1-\varphi)y \cdot (1-\varphi)^{w_b/w_q} = y(1-\varphi)^{1/w_q}$ since $1 + w_b/w_q = 1/w_q$.) The sell preserves L at $\varepsilon = 0$, so $L_s = (1-\varphi)L$.

2.4 EMA band and LTV (feasibility only)

Open rejects if post-trade spot exits EMA$\cdot[1-\beta, 1+\beta]$ on the trade's side (`ensure_open_ema_band`). Short LTV requires $C \geq \lambda \cdot$ (debt mark). Side capacity caps φy (lift TAO). These bound reachable φ; they do not appear inside the pool algebra of a feasible open.

2.5 Readd

Custody returns $(\varphi y, P) = (\varphi y, \varphi x)$.

- **Buggy** (pre-fix): `adjust_protocol_liquidity`→ `update_weights_for_added_liquidity` at the *current* reserves (x_c, y_c):

$$w_q' = \frac{w_q\, x_c\,(y_c + \varphi y)}{w_b\, y_c\,(x_c + \varphi x) + w_q\, x_c\,(y_c + \varphi y)}, \qquad w_b' = 1 - w_q'. \tag{4}$$

  Spot is held at the post-pump mark; L generally jumps.

- **Fixed** (design (a)): credit $(\varphi y, \varphi x)$ with (w_b, w_q) untouched.

3 Cycle: short → pump → wallet repay → unstake

Cycle 1 (Two-wallet drain shape (clone a13))**.** Attacker posts collateral C, opens a short of lift fraction φ, receives escrow E. A pumper buys alpha with S TAO. Attacker buys B TAO of alpha (enough to cover P), closes with `repay_from_wallet`, then both unstake residual alpha.

3.1 Base case $\varepsilon = 0$

1. **Open.** Lift φ, force-sell $P = \varphi x$: state (x, y_s) with y_s as in (3); escrow E; $L_s = (1-\varphi)L$; weights unchanged.

2. **Pump S.** Buy with S via (2) at $\varepsilon = 0$:

$$y_p = y_s + S, \qquad x_p = x\left(\frac{y_s}{y_p}\right)^{w_q/w_b}, \qquad \alpha_{\text{pump}} = x - x_p.$$

   L unchanged ($L_p = L_s$).

3. **Buy B for repay.** $y_b = y_p + B$, $x_b = x_p \cdot (y_p/y_b)^{w_q/w_b}$, $\alpha_{\text{buy}} = x_p - x_b$ (need $\alpha_{\text{buy}} \geq P$).

4. **Wallet repay + readd.** Burn P alpha from wallet; refund $E + C$; credit $(\varphi y, P)$ into (x_b, y_b).

   - Buggy: apply (4); get (w'_b, w'_q), reserves $(x_b{+}\varphi x,\ y_b{+}\varphi y)$, new $L' = (x_b{+}\varphi x)^{w'_b}(y_b{+}\varphi y)^{w'_q}$.
   - Keep: same reserve credit, weights stay (w_b, w_q).

5. **Unstake** residual alpha $\alpha_{\text{pump}} + (\alpha_{\text{buy}} - P)$ via (1) at the post-readd weights.

3.2 Attacker PnL

Combined free of attacker + pumper (collateral is internal):

$$\Delta = -S - B + E + \underbrace{\Delta y_{\text{unstake}}}_{\text{sell residual } \alpha}. \tag{5}$$

When the residual sell restores x near baseline, empirically $\Delta \approx y_0 - y_{\text{end}} = -\Delta\texttt{SubnetTAO}$. Profit iff $\Delta > 0$.

3.3 Production fee

Replace every swap with (1)–(2) at $\varepsilon = 33/65535$. Lift / readd / weight update unchanged. No fee is charged on lift or on the weight update itself.

4 Prediction of the SN29 clone

Inputs from the clone report (tip `4d3dc1585`). Script: `scripts/no_arbitrage_derive.py`.

4.1 Two-wallet (a13)

$y = 7449.956$, $x = 3\,153\,861$, $w_q = 0.471657$, $C = 1500$, $S = 9000$, $B = 2363$. Infer φ from $y_s = 6758.050$ via (3):

$$\varphi = 1 - (y_s/y)^{w_q} \approx 0.04493.$$

model	Δ free	Δ SubnetTAO	w_q after readd
report (clone)	+289.269	−297.594	0.450646
pallet math, $\varepsilon = 0$, buggy	+297.815	−297.815	0.450638
pallet math, $\varepsilon = 33/65535$, buggy	+289.304	−295.026	0.450646
pallet math, $\varepsilon = 33/65535$, keep-w	−281.964	+276.242	0.471657

Production-fee buggy path matches the clone free PnL to $< 0.05\,\tau$. Keep-weights flips the sign.

4.2 Single-wallet (a15)

$y = 7152.475$, $w_q = 0.450710$, $C = 1200$, $S = 7000$, buyback close. Fit $\varphi \approx 0.033$ (LTV/capacity-feasible).

model	Δ free	Δ SubnetTAO	w_q after
report	+121.699	−127.946	0.438945
$\varepsilon = 0$, buggy	+121.794	−121.794	0.439474
$\varepsilon = 33/65535$, buggy	+115.482	−119.540	0.439478
$\varepsilon = 0$, keep-w	−117.798	+117.798	0.450710

Theorem 1 (Drain = weight re-derivation on readd)**.** *On the observed SN29 states, the open→pump→close→unstake cycle has $\Delta > 0$ if and only if readd uses* (4) *at the post-pump mark. With weights held fixed, $\Delta < 0$ (fee-lossy). Fees are a small correction; the base-case ($\varepsilon = 0$) buggy path is already profitable.*

Remark 2 (Theorem status). Theorem 1 is a closed-form / numeric identity on the observed SN29 clone books. A general closed-form $\Delta \leq 0$ for *every* keep-weights cycle is still open. Section 6 supplies the operational pass/fail surrogate used for shipping.

5 Root fix

`readd_protocol_liquidity` credits $(\Delta y, \Delta x)$ **without** calling (4). Emissions still use `adjust_protocol_liquidit` Stake/unstake unchanged.

This is not "Uniswap v2": $w_q \neq \frac{1}{2}$ stays a weighted Balancer CPMM; only lending readd skips asymmetric-join reweight (Section 9).

6 Pass/fail criterion (keep-weights)

Definition 1 (Atomic action)**.** An *atomic action* is one of: `open_short`, `open_long`, `buy/AddStake`, `sell/RemoveStake`, `close` (wallet repay or escrow buyback). Force-sell / force-buy inside open, and protocol readd on close, are not separate attacker actions. A k-way equal split of one buy or sell counts as k actions.

Definition 2 (Self-contained sequence)**.** A sequence is *self-contained* if it starts and ends with no open loan on the subnet and with all attacker-controlled alpha inventory unwound to TAO. Multi-coldkey means the attacker may use several coldkeys (e.g. a dedicated pumper); attacker net TAO Δ is the sum of free-balance changes across those keys.

Definition 3 (Side-cap $\varphi_{\max}$)**.** With no open book, $G = y$. Default `LiftParams` give

$$\mathrm{Cap} = \max(f \cdot G,\ \min(G - y^\star,\ 0.5\,G)), \qquad \mathrm{Cap}_{\mathrm{short}} = 0.5 \cdot \mathrm{Cap},$$

where $f = 0.33$ if $G \geq 3000\,\tau$ else $f = 0.15$, and $y^\star = \mathrm{ref}/((1.10)^{w_b} - 1)$. Short $\varphi_{\max} = \mathrm{Cap}_{\mathrm{short}}/y$; long $\varphi_{\max} = \mathrm{Cap}_{\mathrm{long}}/y$.

Definition 4 (Bounded-search pass/fail)**.** Fix $N \in \mathbb{N}$ (default $N = 8$) and production fee $\varepsilon = 33/65535$. On each live reserve snapshot, under **keep-weights** readd:

PASS $\iff \nexists$ self-contained sequence of length $\leq N$ with $\varphi \in (0, \varphi_{\max}]$ and attacker $\Delta > 0$.

The search **must** include multi-coldkey and trade-split variants. The same grid run against buggy readd must find $\Delta > 0$ on at least one cell (sanity that the harness detects the known drain).

Search status. Script `scripts/no_arbitrage_cycle_search.py` on live finney snapshots (SN29, SN64 Chutes, SN51, SN4, SN119 thin) with $N = 8$, up to 3-way splits, multi-coldkey on/off, short and long, wallet and buyback closes, φ up to each side-cap ($\varphi_{\max} \in \{0.165, 0.25, 0.25, 0.25, 0.075\}$): keep-weights reports **PASS** on 20 087 live cells (max $\Delta \approx -0.01\,\tau$); buggy is profitable on 16 619 cells of the same grid (sanity). JSON: `no_arbitrage_cycle_search.json`. Closed-form coverage beyond Theorem 1 remains open.

7 Deliberate volatility: slippage $\approx 1/(1-\varphi)$

A short lift removes $(\varphi x, \varphi y)$ into custody at unchanged spot, leaving executable reserves $((1-\varphi)x, (1-\varphi)y)$. (The subsequent force-sell then moves the mark to $y_s = y(1-\varphi)^{1/w_q}$ at $\varepsilon = 0$; that is a price move, not the depth claim.) Relative to the un-lifted book, a small trade of fixed TAO size therefore sees fractional price impact scaled by

$$M(\varphi) \;\approx\; \frac{1}{1-\varphi}. \tag{6}$$

This is intentional: lending lifts liquidity out of the AMM for the life of the loan and makes the pool shallower. Figure 1 plots $M(\varphi)$ up to each subnet's $\varphi_{\max}$ (side-cap), with the cap marked.

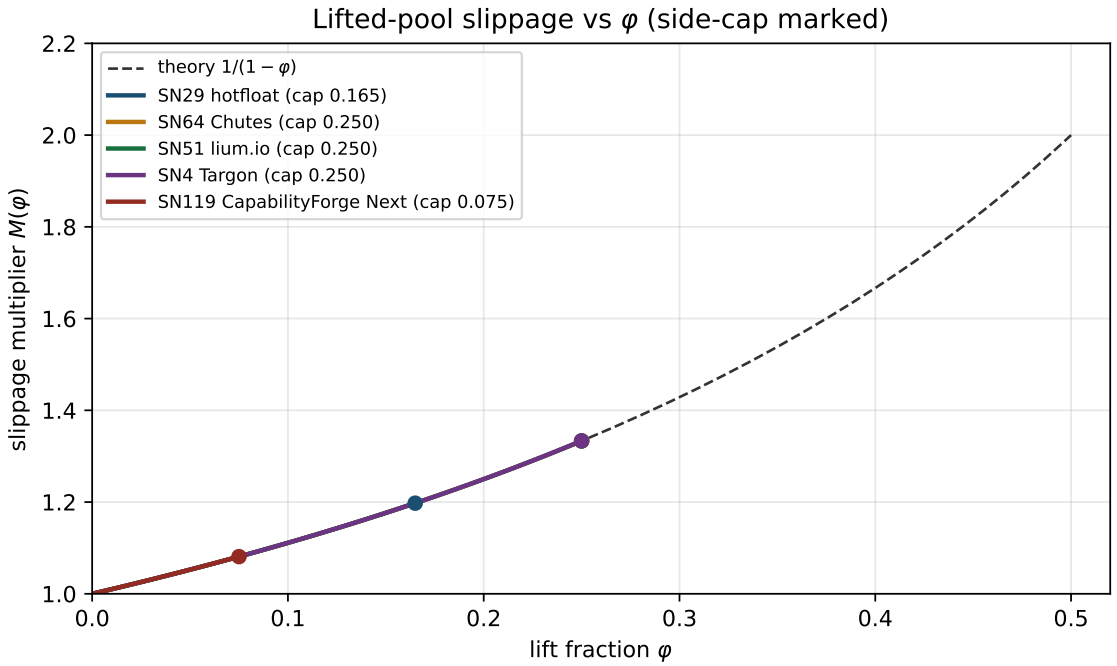


Figure 1: Slippage multiplier $M(\varphi) = 1/(1-\varphi)$ vs lift fraction for live SN29, SN64 (Chutes), SN51, SN4 and thin SN119, each truncated at its short side-cap $\varphi_{\max} = \mathrm{Cap}_{\mathrm{short}}/y$. Script: `no_arbitrage_slippage_chart.py`.

8 Cross-check: μ, September 2026

μ's note (September **2026**, not 2025) writes a pool factor $F(f_h, \varphi, w_2)$ and states fees are ignored. After the derivation above we can map without guessing:

μ symbol	Our symbol	Reason
φ	φ (lift fraction)	identity $y_s/y = (1-\varphi)^{1/w_q}$ at $\varepsilon = 0$ is exactly μ's fa
w_2	w_q	quote weight in that identity
fee	$\varepsilon = 0$	μ: fees ignored
f_h	size / haircut parameter in μ's extended model	**not used** in our SN29 prediction; we size with obse

Remark 3. We do **not** import μ's closed form as an assumption. Equation (3) is proved from lift + force-sell in Section 2. That it coincides with μ's $(1-\varphi)^{1/w_2}$ term is the cross-check. Our profitability statement is $\Delta > 0$ in (5) under buggy readd — an attacker convention — whereas μ's $F < 1$ is a pool-cost convention on a fee-free short cycle without the weight-ratchet term. The SN29 drain is the ratchet (4), which that September fee-free F does not encode.

9 Fixed weights are not Uniswap v2

Main re-derives weights on *asymmetric liquidity adds* (emissions), so those adds do not move spot. Swaps never re-derived weights. Uni v2 is the special case $w_b = w_q = \frac{1}{2}$. Holding $w_q \approx 0.47$ fixed is still weighted Balancer. Design (a) loses only price-neutrality of *lending readd after skew*; that is the intentional root fix.

Recommendation (Jacob). Ship (a). Optional hardening: restore ΔL by a current-ratio proportional join and mark-to-market the custody residual (design (b)). Do not restore current-mark (4) on lending readd.

References

[1] μ, September 2026 hand derivation of the fee-free short-cycle factor $F(f_h, \varphi, w_2)$. φ = lift fraction; fees ignored. Cited as cross-check only (Section 8).

[2] *Lift-based lending on main's Balancer pools*, v5.4, `pallets/lending/docs/lift-lending-spec.md`.

[3] SN29 drain reproduction on clone tip `4d3dc1585` (two-wallet $+289\,\tau$, single-wallet $+122\,\tau$).